

Predictive Modeling Preliminary Results

DSE6111

JUNE LEMIEUX

This preliminary report briefly refreshes key information first identified in the Analytic Plan report previously submitted. Additionally, this report will identify the models under review, the initial findings and challenges. Finally, the report will identify future opportunities for processing and refinement.

ABC Corporation seeks to identify the risk of customer attrition and to predict which customers are most at risk of leaving. To accomplish this, an Attrition_Flag response will be developed and predicted using at least two predictive models. Further, the company will be provided with a list of attributes that highly influence the risk of attrition. The company can develop marketing initiatives with these attributes in mind and target customers accordingly.

ABC Corporation has provided customer data to build two models: one logistic regression model and one kNN model. To start, preprocessing and setup steps were completed to support each model. In the customer data, one feature of the data is each customer's status: either Attrited or Existing. This flag was changed from text to numeric, with Attrited Customer becoming a 1 and an Existing Customer becoming a 0. This will assist with prediction for both models. Additionally, the "ClientNum" identifier was removed from the dataset, as it does not add any value to the predictions. The full data set was split into training and testing data sets, with 75% of the data carved out to the training set, and the remainder going to the testing set.

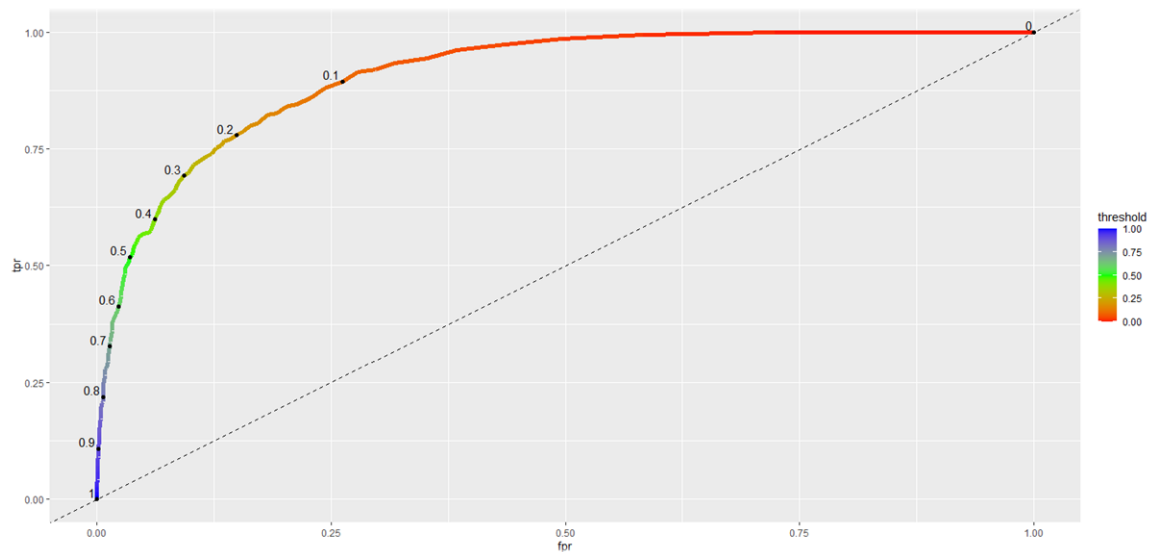
For both models, the following features were used:

- Months_on_book
- Total_Relationship_Count
- Months_Inactive_12_mon
- Contacts_Count_12_mon
- Total_Amt_Chng_Q4_Q1
- Total_Trans_Amt
- Total_Trans_Ct
- Total_Ct_Chng_Q4_Q1
- Avg_Utilization_Ratio

Additionally, recipes were used to organize the work. Although no nominal predictors were used at this point, the recipes were configured to create dummy variables as needed, and the predictors were scaled and centered.

For the logistic model, the generalized linear model engine was used with logistic regression. When a risk prediction was greater than .5, it was assigned as "At Risk". In the testing set of 2,532 records, the model identified 287 of those test customers at risk of attrition, or approximately 11% of the customer base. Additionally, the model correctly identified the at-risk customers and not-at-risk customers approximately 89% of the time. The ROC curve is presented below. By my estimation, the model performs better than the null model, as the model identifies that additional predictors are needed to

sufficiently estimate the Attrition_Flag. (James G., 2023)



For the kNN model, five cross validation folds for tuning were created. The knn engine was used with a classification mode. The tuning grid had five neighbors: 5, 10, 15, 20, and 25). The tuned model identified 10 neighbors as the least complex but most accurate number of neighbors. The related roc_auc metric was just slightly lower than other roc_auc metrics. Like the logistic model, the kNN model performs better than the null model, as the model identifies that additional predictors are needed to sufficiently estimate the Attrition_Flag. (James G., 2023) (Max Kuhn, n.d.) (Svancer, 2023)

[Student edit: I was unable to successfully get an ROC plot to work on the kNN model despite trying for an embarrassingly long amount of time. I endeavor to improve on this for the final report] (Patrick A. Ward, 2024)

For the next phase of the project, the effect of adding more predictors to each model will be reviewed. Additionally, these models and others (including naïve Bayes, random forest, and a boosting algorithm) will be compared for accuracy and performance. One model will be selected to estimate the Attrition_Flag risk to address ABC Corporation's needs. Also, the final report will identify the customer attributes with the most influence on attrition risk. The model and attribute analysis results will be explained in the context of actionable next steps that ABC Corporation can take to reduce customer attrition.

References

James G., W. D. (2023). *An Introduction to Statistical Learning with Applications in R.* : Springer.

Max Kuhn, D. V. (n.d.). *K-nearest neighbors via kkn*. Retrieved from Parsnip:
https://parsnip.tidymodels.org/reference/details_nearest_neighbor_kknn.html

Patrick A. Ward, P. (2024, September 30). *K-Nearest Neighbor: {tidymodels} tutorial*. Retrieved from Optimum Sports Performance: <https://optimumsportsperformance.com/blog/k-nearest-neighbor-tidymodels-tutorial/>

Svancer, D. (2023). *Machine Learning with tidymodels*. Retrieved from George Mason University School of Business: https://gmubusinessanalytics.netlify.app/model-fitting-examples.html#KNN_Classification